

From centroid to entrance: a global assessment of POI access locations for accessibility

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1 Introduction

Points of interest (POIs) and residential locations serve as origins and destinations when estimating travel demand and modelling behaviour. The underlying data come from travel surveys, property registries, GPS traces and, increasingly, OSM, which acts both as a crowdsourced database and as an openly licensed container for license-compatible authoritative datasets. When footprints are available, most researchers approximate each POI by the centroid of its polygon (or the parcel where no building is mapped). On small single-purpose buildings this is acceptable; on larger and multi-building sites the centroid can misrepresent the access point by hundreds of metres—sometimes kilometres—which is not negligible for walking or first-/last-mile transit.

Entrances can be tagged explicitly via OSM `entrance=*` (`main`, `customers`, `home`, `yes`, ...) but remain sparsely mapped outside a few well-covered regions. Large sites usually expose several entrances with distinct functions—main, visitors, employees, deliveries, emergency—each with its own accessibility profile. Precise access points matter most for paratransit and wheelchair users, transit riders for whom the stop-to-door segment dominates total travel time, cyclists seeking secure parking, and routing engines including autonomous taxis, for which curbside drop-off precision is a safety and usability requirement.

A growing literature characterises the completeness and positional accuracy of OSM buildings and POIs (Haklay, 2010; Girres & Touya, 2010; Fan et al., 2014; Herfort et al., 2023; Klinkhardt et al., 2023) and shows that entrance-level detail changes transit catchments (Gutiérrez & García-Palomares, 2008; El-Geneidy et al., 2014), wheelchair routing (Neis & Zielstra, 2014), and curbside outcomes for autonomous vehicles (Hunter et al., 2024; Wang et al., 2025). The `entrance=*` tag itself remains a blind spot: Hu et al. (2021) found only ≈ 60 tagged buildings in their entire London extract. Our prior work (Bourbonnais et al., 2026) validated 1,034 POIs from five Québec household travel surveys and observed median positional errors around 20 m across OSM, Overture Maps, the Québec Property Registry and Google Places, with long tails dominated by

university campuses and hospital complexes; it explicitly flagged the centroid-vs-entrance comparison as a future research question. To our knowledge, no study yet quantifies `entrance=*` coverage at scale, nor the cost of substituting a centroid for an entrance. We quantify that gap on a worldwide random sample of POIs by comparing centroid baselines to the nearest tagged entrance, and show how entrances can be mapped to support walking, cycling, transit and autonomous-vehicle use cases.

2 Methodology

The study is a global exploratory analysis. POIs are coded along four axes: building footprint present or not; single- vs. multi-building site; `entrance=*` present or not; and specificity of the entrance tag (values other than `yes`). Coverage extends beyond buildings to polygon-mapped sites without a unique footprint—e.g. university campuses, large factory areas, or sport pitches—where the area polygon itself supplies the centroid baseline.

Two complementary sampling strategies. A global pass tessellates the world into 10×10 km cells over population density using GHS-POP and built volume using GHS-BUILT-V (Joint Research Centre, European Commission) and draws cells with probability proportional to a *blended* mix of population and built volume (default $\alpha = 0.5$). The built-volume term prevents low-residency but high-activity fabric—campuses, industrial parks, transport hubs—from being under-sampled by a purely population-weighted draw. The current round is purely random, which under-samples the large multi-building sites where the centroid proxy is most misleading. A complementary targeted pass—the analyst supplies a precise point or OSM node/way/relation—will characterise the upper tail on a curated list of large sites; both passes share the same pipeline.

Reference entrances and centroid baselines. Reference entrances are established from open and proprietary street-level imagery (Mapillary, KartaView, Panoramax, Google Street View, and the Chinese Baidu Panorama and AMap services) and, where available, architectural or wayfinding plans. A non-trivial share of randomly selected POIs are rejected at this stage—most often because aerial and street-level imagery do not let the analyst confidently locate entrances, or because the underlying tag is obsolete (e.g. a closed shop). Four centroid baselines are compared against each OSM entrance: parcel, main-building, and all-buildings centroids where a footprint exists, and the area-polygon centroid for sites without a unique footprint (campuses, factory areas, sport pitches).

Distance metrics and persisted measurements. The analyst draws a polyline on the focus map—a single Euclidean segment, or a multi-vertex trace along sidewalks/desire lines where mapped paths are missing. Each polyline is persisted with its geodesic length, walking duration at a configurable speed,

coordinates, an analyst-supplied *purpose* (nearest entrance, main entrance, transit stop, off-street parking, driving road, and walking/cycling/driving network combinations) and *entrance class* (**main**, **customers**, **home**, **emergency**, service-employees, service-delivery, **garage**, or one of four centroid baselines: parcel, main-building, all-buildings, or area).

Pipeline and OSM contributions. The pipeline is a Rust backend over PostgreSQL/PostGIS, fed by Overpass for OSM features, with a React/MapLibre review interface; all code, the sampled POI list and computed metrics are released under an open licence. OSM edits made during the analysis to add missing entrances and pedestrian-access geometry (footpaths and connecting paths) for the selected POIs carry the `#StateOfTheMap2026` and `#EntranceAnalysis` hashtags in their changeset comments, so every contribution is traceable. Repository: <https://github.com/chairemobilite/stateofthemap2026> (translations, drafting and parts of the pipeline were assisted by Anthropic Claude Opus 4.6 and 4.7; all data, methods and findings remain the authors’).

3 Review interface

The review service is a single-page Rust/React/MapLibre application (Figure 1) with four screens. *Sampling* draws a candidate 10×10 km cell or accepts a custom anchor and previews its weights. *Kept bboxes* lists accepted cells and their POI inventory. *POI focus* centres the map on a selected POI, lets the analyst draw, classify and persist measurements (purpose $\times$ entrance class) over OSM buildings and entrances inside a configurable focus radius, and provides right-click deeplinks to the imagery providers listed above and to the OSM iD editor. *Stats* aggregates min, max, mean and median length and walking duration over all persisted measurements, grouped pairwise by `measurement_type`, `entrance_type`, and `start_origin`.

4 Preliminary findings

The initial pass is a 100% random draw of ≈ 25 POIs, predominantly in North America (Figure 2), yielding 140 measurements. For each POI we pair the centroid-anchored polyline with the nearest tagged-entrance polyline drawn to the same target and report the per-POI difference; a positive value means the centroid proxy yields a longer walk than the real entrance. We aggregate by target family across all centroid types (area, main-building, multi-building).

Across the 25 paired observations, 24 have longer walk from centroid to nearest road network than from entrance (single exception where the centroid is closer than the entrance). The pooled difference is +16 m median, +80 m max for distance to nearest road network ($n=15$); +37 m median, +92 m max for distance to nearest transit stop ($n=10$).

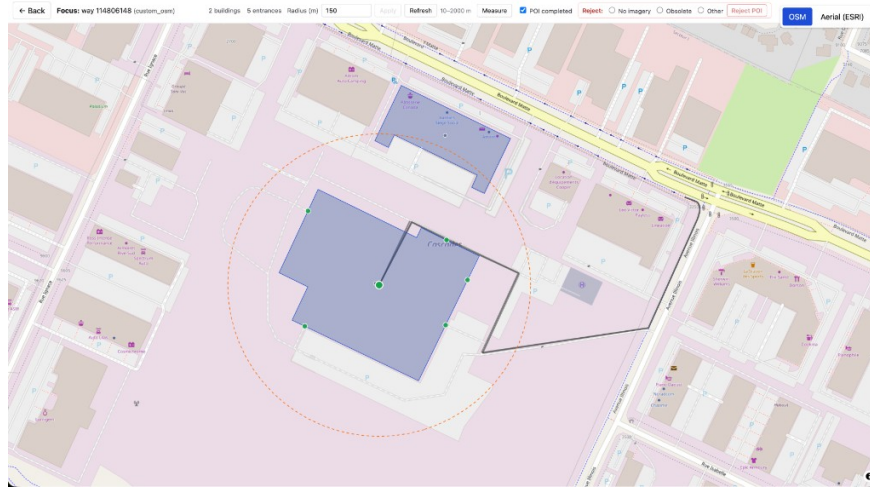


Figure 1: POI focus screen for one POI: building polygons, entrance nodes (green), the configurable focus radius (orange), and an analyst-drawn polyline from the building centroid to the nearest tagged entrance.

These are preliminary figures from a small random draw, not final results. The full paper will combine a substantially larger random sample, a targeted pass on large multi-building and area-coded sites, and the regional Québec field-validated analysis, all through the same pipeline.

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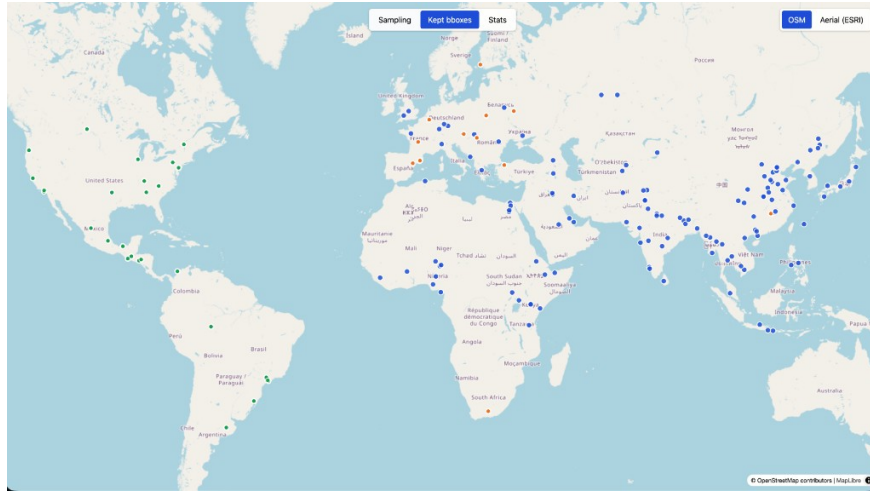


Figure 2: Random initial sample: kept 10×10 km cells worldwide (green: completed POI; orange: in progress; blue: not yet started).

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